

GANYUSHKINO, Kazakhstan

WMO: 347980

Lat: 46.6053N

Lon: 49.2381E

Elev: -75

StdP: 14.74

Time Zone: 5.00 (E05)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.4	-0.2	-11.6	2.9	-5.4	-5.8	4.0	0.4	31.9	23.3	30.9	25.3	6.1	90	0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.5	96.9	69.6	94.4	69.1	91.7	68.5	74.1	88.0	72.3	87.1	70.8	86.0	10.4	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
69.8	109.2	82.0	67.4	100.3	79.7	65.4	93.4	78.9	37.6	87.9	35.9	87.3	34.6	85.9	83.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	(q)
27.9	24.8	20.8	DB	-10.8	101.2	8.0	2.9	-16.6	103.2	-21.2	104.9	-25.7	106.5	-31.5	108.6	
			WB	-10.9	77.3	7.8	2.8	-16.5	79.3	-21.0	81.0	-25.4	82.6	-31.0	84.6	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	51.1	22.1	23.3	37.0	51.8	66.4	75.1	78.9	76.5	64.6	50.5	37.1	27.7
	DBStd	21.86	11.04	10.74	7.52	7.54	7.13	6.03	4.98	6.28	7.08	7.87	9.06	9.49
	HDD50	3260	864	748	405	66	1	0	0	0	2	91	391	692
	HDD65	6373	1329	1168	867	398	68	5	0	4	90	450	837	1157
	CDD50	3654	0	0	3	120	509	752	897	823	438	107	5	0
	CDD65	1292	0	0	0	2	111	307	432	362	77	1	0	0
	CDH74	14423	0	0	0	49	1050	3282	5103	4198	730	11	0	0
	CDH80	6677	0	0	0	3	322	1481	2568	2108	195	0	0	0
	WSAvg	9.6	10.0	10.4	11.3	11.2	10.0	9.4	8.6	8.1	8.9	8.9	8.6	9.8
	PrecAvg	6.3	0.4	0.3	0.4	0.6	0.6	0.7	0.6	0.5	0.6	0.5	0.5	0.4
Precipitation	PrecMax	10.2	1.1	0.7	1.3	1.6	2.2	1.5	2.7	2.5	2.6	2.0	1.3	1.0
	PrecMin	3.5	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
	PrecStd	1.9	0.3	0.2	0.3	0.5	0.6	0.5	0.6	0.6	0.6	0.5	0.4	0.2
	PrecStd	1.9	0.3	0.2	0.3	0.5	0.6	0.5	0.6	0.6	0.6	0.5	0.4	0.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4% DB	43.2	46.3	61.2	79.4	90.9	98.8	99.8	100.4	90.9	75.5	60.8	48.1	
	0.4% MCWB	39.9	41.1	50.1	58.7	65.2	68.7	70.3	69.8	67.4	60.0	51.1	43.8	
	2% DB	39.5	41.9	55.9	74.3	86.8	95.0	96.7	96.8	85.0	71.0	56.1	44.4	
	2% MCWB	37.0	37.7	46.6	57.2	64.9	68.7	70.4	69.1	65.0	56.9	50.1	41.0	
	5% DB	36.8	38.9	51.8	69.9	83.7	91.6	94.3	93.9	81.4	67.2	52.4	41.4	
	5% MCWB	34.7	35.4	43.5	55.0	63.4	67.6	69.7	69.0	63.5	55.4	47.7	38.9	
	10% DB	34.6	36.0	48.2	65.6	80.3	88.3	91.7	90.8	78.0	63.1	48.8	38.3	
	10% MCWB	32.7	33.2	41.7	52.7	62.2	66.4	69.1	68.3	62.3	53.5	45.2	36.0	
	0.4% WB	40.5	42.4	51.0	60.2	69.8	74.5	77.4	75.4	70.4	62.3	53.8	44.9	
	0.4% MCDB	42.4	45.4	59.9	75.5	83.7	89.1	90.9	88.7	83.3	72.4	58.3	47.3	
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2% WB	37.2	38.6	47.2	57.8	67.3	71.7	75.0	73.2	68.0	59.2	50.9	41.5	
	2% MCDB	39.1	41.4	54.9	71.7	81.4	87.6	88.4	88.6	79.7	67.3	55.0	43.8	
	5% WB	34.8	35.7	44.5	56.0	65.7	70.1	73.0	71.5	65.9	56.5	48.0	38.9	
	5% MCDB	36.7	38.3	50.8	67.9	79.6	85.6	86.9	87.2	77.9	64.2	51.9	41.3	
	10% WB	32.8	33.5	42.0	53.7	64.0	68.4	71.3	70.0	63.8	54.4	45.2	36.2	
	10% MCDB	34.5	35.8	47.4	63.8	77.0	83.6	86.1	85.9	75.6	61.9	48.8	38.1	
Mean Daily Temperature Range	MDBR	10.8	13.2	14.7	19.4	18.9	20.0	21.5	23.8	22.0	18.6	13.3	9.3	
	5% DB	10.7	13.9	20.6	25.2	22.6	25.2	25.3	27.0	25.6	23.7	17.7	10.3	
	5% WB	9.2	10.4	12.7	11.4	8.7	9.0	9.2	9.6	10.9	12.5	12.2	8.8	
	MCDBR	10.6	12.9	18.5	23.9	19.9	22.5	21.6	24.2	22.3	20.7	16.6	9.8	
	MCWBR	9.3	10.3	12.0	11.7	9.4	9.2	9.6	10.0	10.9	12.7	12.0	8.7	
	MCWBR	9.3	10.3	12.0	11.7	9.4	9.2	9.6	10.0	10.9	12.7	12.0	8.7	
Clear-Sky Solar Irradiance	taub	0.362	0.384	0.427	0.497	0.473	0.447	0.452	0.456	0.425	0.394	0.371	0.374	
	taud	2.113	2.066	2.049	1.921	2.062	2.196	2.194	2.156	2.182	2.217	2.190	2.097	
	Ebn at Noon	216	239	247	241	253	261	258	251	247	234	213	193	
	Edh at Noon	30	38	45	56	51	45	44	44	39	33	28	28	
All-Sky Solar Radiation	RadAvg	345	641	1070	1577	1981	2183	2106	1878	1434	878	449	259	
	RadStd	54	76	112	131	157	155	100	92	95	85	66	44	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (2 neighbors)	+0.95	N/A	N/A	+2.63	N/A	N/A	N/A	N/A	+306	+207

Nomenclature: See separate page